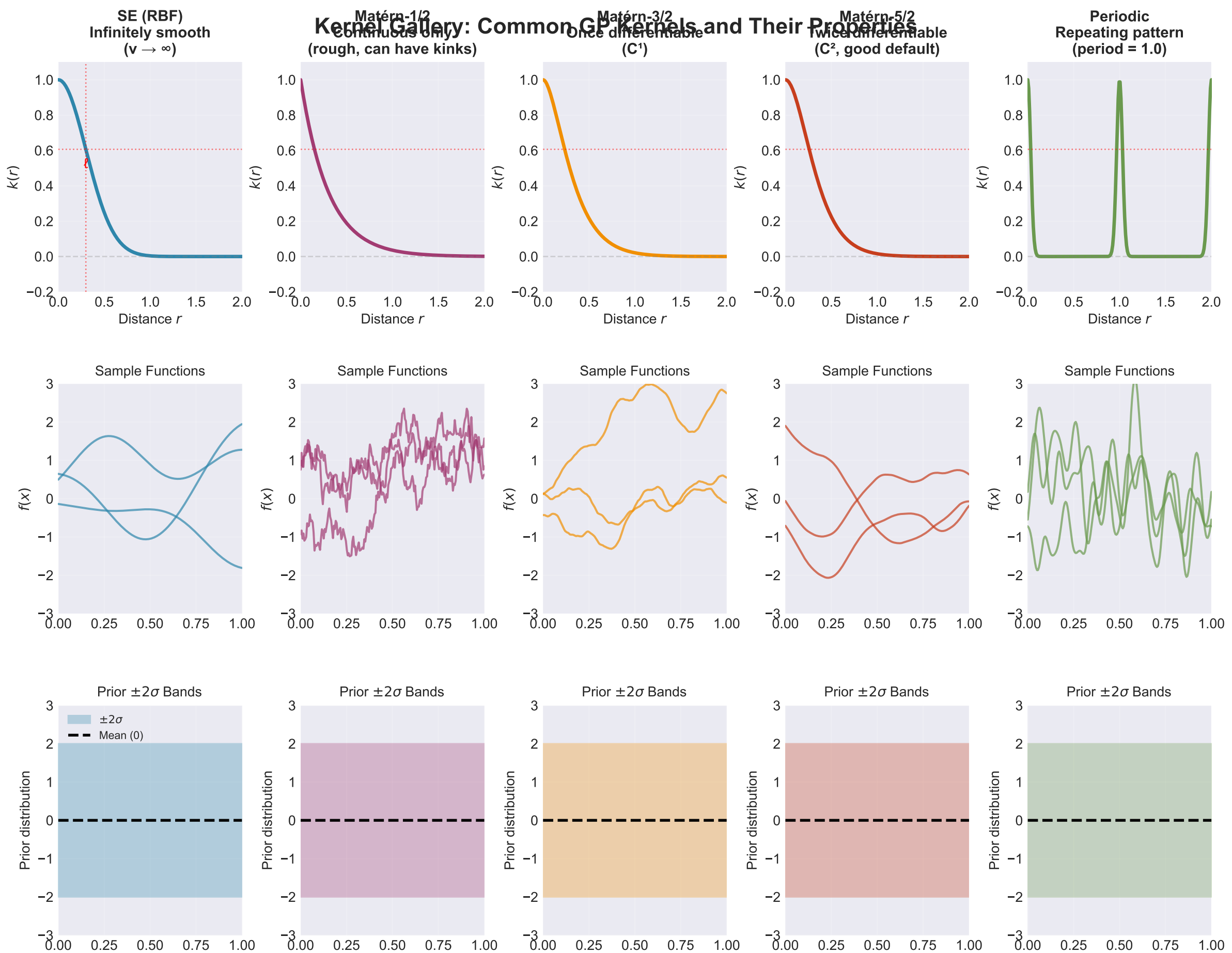


Kernel Gallery: Common GP Kernels and Their Properties



Top row: Kernel correlation $k(r)$ vs distance r . Middle row: Random function samples showing smoothness. Bottom row: Prior confidence bands. SE is infinitely smooth (wide, wiggly). Matérn-1/2 is rough (can have kinks). Matérn-5/2 balances smoothness with realism (recommended default). Periodic captures repeating patterns.